# On Market Making for a New Exchange: EUR/USD Designated Market Maker on Exchange A

Edouard Blanc  Clément Dardare  Julien Simeoni

Paris Dauphine University — MSc 203

April 2026

## Contents

**1 Introduction** **3**
1.1 Motivation . . . 3
1.2 Objective . . . 3
1.3 Simulation Environment . . . 3
1.4 Evaluation . . . 4
1.5 Report Structure . . . 4

**2 Order Book** **5**
2.1 Data structure . . . 5
2.2 Submissions and cancellations . . . 5
2.3 Matching rule . . . 5
2.4 Fills as events . . . 6

**3 Market Simulator for Venues B and C** **7**
3.1 Noised mid . . . 7
3.2 Spread regimes . . . 7
3.3 Depth . . . 8
3.4 One stock, two venues . . . 8
3.5 Validation . . . 8

**4 Market-Making Logic on Exchange A** **10**
4.1 Fair mid under latency asymmetry . . . 10
4.2 Reservation price: Avellaneda–Stoikov with infinite horizon . . . 10
4.3 Three-component spread with fee floor . . . 10
4.4 Asymmetric skew and ladder . . . 11
4.5 Selective re-quoting . . . 11
4.6 Hedge router . . . 12

**5 Client Order Flow on Exchange A** **13**
5.1 Poisson arrivals with price-sensitive intensity . . . 13
5.2 Market-versus-limit mix and limit placement . . . 13
5.3 Heavy-tailed order sizes . . . 14
5.4 Consumption by the order book . . . 15

**6 Inventory and P&L Tracking** **16**
6.1 Sign and fee convention . . . 16
6.2 Realized versus mark-to-market . . . 16
6.3 Inception spread versus mid-to-mid revaluation . . . 16
6.4 Continuous MtM and final inventory . . . 17

**7 Phase 1 Backtest Report** **18**
7.1 Simulation setup . . . 18
7.2 Aggregate results . . . 18
7.3 Market prices — three venues (Subject 7.1–7.2) . . . 18
7.4 Top 10 MM fills (Subject 7.2) . . . 19
7.5 Inventory vs. EUR/USD (Subject 7.3) . . . 19
7.6 Per-trade MtM P&L percentiles (Subject 7.4) . . . 19
7.7 Fill-rate analysis (Subject 7.5) . . . 20
7.8 Interpretation . . . 21
7.9 Take-away for Phase 2 . . . 22

**8 Phase 2 — Calibration from Trading History** **23**

**9 Phase 3 — Fallback Liquidity Under HFT Competition** **24**

**10 Conclusion** **25**

# 1 Introduction

## 1.1 Motivation

A newly launched exchange — referred to throughout this report as exchange $A$ — has mandated our firm as the sole Designated Market Maker for its EUR/USD spot book. Two established venues, $B$ and $C$, remain the primary sites of price discovery, carrying on average 75% and 25% of the daily volume respectively. We receive market data from $B$ with a 200 ms delay and from $C$ with a 170 ms delay, while high-frequency participants reach the same data in 50 ms and can act on exchange $A$ before our own view of fair value has refreshed. The mandate commits us to continuously quoting ten levels on each side of the book at a common 1 bp tick, with a starting capital $K$ split evenly between EUR and USD and an asymmetric fee grid. A delta limit caps the EUR inventory we are willing to warehouse before externalising risk as taker flow on $B$ and $C$.

Three constraints shape every design decision that follows. First, on the day exchange $A$ goes live no historical fills or counterparty behaviour are observable, so the initial quoting policy cannot be learnt from data and must be derived from first-principles microstructure models. Second, the latency gap to HFTs exposes every resting quote to adverse selection, which the pricing model must compensate for rather than ignore. Third, because we cannot market-make on $B$ or $C$, the delta limit turns inventory management into an active cross-venue hedging problem rather than a passive book-balancing exercise.

## 1.2 Objective

Our goal is to deliver an end-to-end market-making system that addresses the three constraints above and can be evaluated under the conditions described in the subject brief. The project is split into three phases matching the mandate timeline. Phase 1 bootstraps liquidity without any trading history, using a heuristic quoting policy derived from classical market-making theory. The fair mid is reconstructed from the two reference venues at their respective latencies, the bid-ask spread is sized to compensate for both inventory risk and adverse selection, and quotes are skewed in the direction of recent order-flow imbalance. A dedicated hedge router externalises excess EUR inventory on $B$ and $C$ whenever the delta limit is approached. Phase 2 re-estimates the Phase 1 parameters by maximum likelihood from a simulated month of fills. Phase 3 studies how our quoting policy should degrade from a price-competitive quoter to a reliable fallback liquidity provider once aggressive HFT market makers populate the inside of the book.

## 1.3 Simulation Environment

Because the exchange has no history at launch and we cannot run the strategy against real counterparties, the evaluation environment is a layered simulation built specifically for this project. A stochastic mid-price generator (GBM, GARCH with jumps, or Heston) produces the fair value underlying both reference venues. A venue simulator wraps that path into a realistic bid/ask and depth profile per venue, with a choice of five spread regimes (static, stochastic, adaptive, asymmetric, and skewed) reflecting the microstructure assumptions we want to stress. A limit order book with a matching engine and fill events sits at the centre of exchange $A$; a Poisson client-flow generator with truncated Pareto sizes delivers counterparty orders whose intensity decays exponentially in

the quoted half-spread; and a reporting layer reconstructs the full mark-to-market and realised P&L decomposition from the trade history.

## 1.4 Evaluation

Each phase is evaluated on the subject-mandated one-month simulated backtest, with the capital $K$ reset to its initial split at the start of every phase. For Phase 1 we report realised and mark-to-market P&L, the percentile envelope of mark-to-market as a function of time since trade inception, top-ten trades by size, full-versus-partial fill rates at each of the ten quoting levels, and inventory trajectories against the delta limit. For Phase 2 we compare the heuristic and calibrated configurations on an independent month using the same diagnostics, so that any difference in performance is attributable to the updated parameters rather than to the market regime. For Phase 3 we measure both our own P&L under HFT competition and the quality of liquidity observed by exchange-$A$ clients, distinguishing small from large trades and quantifying the continuity of two-sided quoting through volatility spikes, one-sided HFT retreats, and venue outages.

## 1.5 Report Structure

The remainder of the report follows the step-by-step structure of the subject. section 2 introduces the order-book data structure and matching rule; section 3 describes the venue simulator and the family of spread regimes supported; section 4 presents the quoting framework in full; section 6 details the inventory and P&L bookkeeping; section 7 gathers the Phase 1 backtest outputs; section 8 analyses the calibration results; section 9 examines the Phase 3 fallback behaviour; and section 10 summarises the findings and outlines the remaining modelling questions.

# 2 Order Book

The order book of exchange $A$ must show a price and a quantity at ten levels on both sides at all times. This section describes the data structure we chose to hold that state, how submissions and cancellations flow through it, and the invariants of the matching engine. The design is exercised on a realistic EUR/USD seed (mid = 1.10, 1 bp tick, ten levels per side, 1M EUR per level).

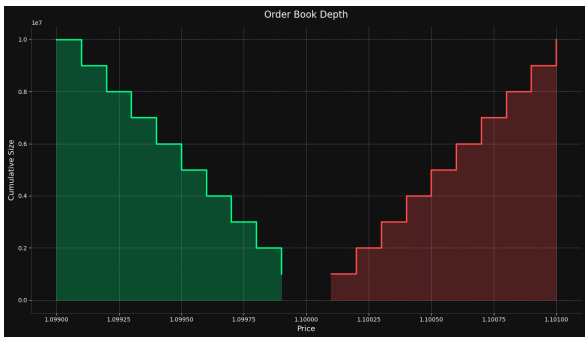
## 2.1 Data structure

Two pressures shape the design. Matching requires a price-time-ordered view of the market-maker (MM) side at clearing time, yet the hot path is dominated by bulk requotes — up to ten cancels followed by ten posts per step. Keeping the book permanently sorted would pay on every mutation for a benefit consumed only once per step.

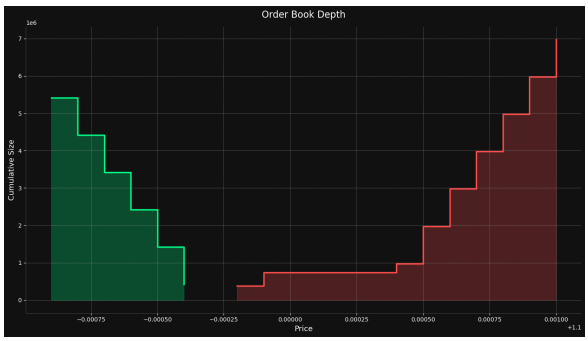
We therefore split orders by origin (MM versus client) into two hash-indexed stores and keep a price-ordered view of the MM side as a lazy cache, rebuilt only when the matcher runs. Cancels and posts touch only the hash stores; the MM side is sorted once per clearing call instead of being re-ordered on every mutation. Time priority within a price level is preserved by a counter that grows by one on every insertion, so later orders carry higher numbers than earlier ones.

## 2.2 Submissions and cancellations

The two sides are handled asymmetrically. MM cancels and posts arrive in bulk and never run the matcher, because passive quotes cannot cross by construction. Client orders are inserted one at a time, and the matcher runs immediately after each insertion.




(a) Seeded book: one order per level, symmetric depth.




(b) After twenty marketable client orders: shallow levels consumed from the top.

Figure 1: Cumulative depth before and after a burst of client flow.

## 2.3 Matching rule

Three invariants govern every clearing call. Client orders only match against MM orders, never against each other: as the sole designated market maker for Phases 1 and 2 we are on one side of every trade. The fill price is always the passive MM price, never the incoming aggressive price — clients pay the spread we quote, which is the only source of gross revenue. And matching follows strict price-time priority with partial fills: the engine walks the sorted MM side, deducts the minimum of the two sizes, logs the match,

and either retires the filled order or leaves the remnant resting. The residual shape in Figure 1 is a direct visual check of this rule: the top of the book is consumed first, deeper levels are untouched.

### 2.4 Fills as events

Every match is dispatched synchronously, as a fill event, to a single registered listener. In the full simulation the listener is the Quoter, which updates inventory, trade history, and the pending top-up queue inside the clearing call. The Quoter therefore never polls the book: the next quote computation already sees the post-fill inventory, which removes staleness bugs on steps that both fill and requote. Decoupling the book from its consumer also keeps the same engine usable in Phase 3, where a separate diagnostic listener records fills for the exchange-client-side liquidity analysis without touching the Phase 1 code path.

# 3 Market Simulator for Venues B and C

Exchanges $B$ and $C$ are external reference venues: we never post on them, we only read them. Our quoting engine on $A$ consumes the best bid, the best ask, and the best-size of each, and nothing else. We therefore do not simulate a full limit order book on $B$ or $C$: all the depth beyond the top of book would be detail the rest of the system never looks at. Instead we simulate a mid-price process and build a synthetic bid/ask/depth layer around it. This is the second of the two options offered by the subject, and it is both cheaper to run and sufficient for everything the market maker on $A$ has to decide. The reduced-form choice keeps realism where it matters — spreads that widen under stress and depth that thins with them — while avoiding the combinatorial cost of simulating individual order arrivals at two venues.

## 3.1 Noised mid

The fundamental value $V_t$ is the GBM (or GARCH, or Heston) path produced by the stock simulator. The observed mid on $B$ and $C$ is $V_t$ plus microstructure noise, in the spirit of Roll [8]:

$$P^{\text{mid}}_{B/C}(t) = V_t + \eta \cdot \sigma \cdot \sqrt{\Delta t} \cdot Z_t, \quad Z_t \sim \mathcal{N}(0,1),$$

with $\eta = 0.1$. Each venue draws an independent $Z_t$ path, so $B$ and $C$ are noised versions of the same fundamental. One subtle point: every spread model below estimates realized volatility from $V_t$, not from $P^{\text{mid}}$, because differencing the noised series would double-count the microstructure variance inside the vol estimator. The zero-mean estimator used for that (see `compute_rv_zero_mean`) is unbiased for zero-drift GBM and lower-variance than the sample std over short windows.

## 3.2 Spread regimes

We benchmarked five regimes of increasing realism — Static, Stochastic, Adaptive, Asymmetric, and Skewed — and kept two. We reject Static because a price-independent width ignores volatility entirely; the market maker is widest exactly when they should be widest, but never adapts. Stochastic scales the width with realized vol but never returns to the competitive floor even in calm markets, which systematically over-charges. Asymmetric mean-reverts realistically but treats up-moves and down-moves the same way, leaving a latency-arbitrage gap for HFTs who, seeing the vol spike 150 ms before us, can infer our quote path. We deploy **Adaptive on** $C$ and **Skewed on** $B$.

**Venue** $C$ — **Adaptive.** A symmetric half-spread proportional to how much realized vol deviates from the long-run input $\sigma$:

$$h(t) \;=\; h_0 \cdot \Big(1 + \alpha \cdot \big(\widehat{RV}_t/\sigma - 1\big)\Big), \qquad h(t) \geq 0.1 \cdot h_0,$$

with $h_0 = \text{mid}(t) \cdot s_{\text{bps}}/10000$ and $\widehat{RV}_t$ an EMA-smoothed rolling vol over a 120 s window. The width is tight at baseline, widens on shocks, and tightens symmetrically when calm returns. A 10% floor prevents collapse near zero. $C$ carries 25% of volume — a smaller venue where informed flow is weaker, competition enforces a tight floor, and the symmetry of up vs. down adjustment is not a material leak.

**Venue $B$ — Skewed.** Same vol-adaptive width, plus a directional shift of the quote centre driven by a short-horizon momentum signal. Let $\hat{m}_t$ be the EMA of log-returns normalised by its steady-state noise floor; the skew is

$$\delta_t \;=\; \gamma \cdot h(t) \cdot \tanh\bigl(\mathrm{sign}(\hat{m}_t) \cdot \max(|\hat{m}_t| - \tau,\, 0)\bigr),$$

with $\tau = 1.5$ the dead-band threshold. Small random wiggles map to $\delta_t = 0$ exactly; only sustained directional moves shift the centre. Posted quotes become ask $=$ mid $+ h + \delta$ and bid $=$ mid $- h + \delta$: width is unchanged, the centre moves in the direction of the trend. $B$ carries the dominant 75% of volume, where short-horizon order-flow imbalance predicts returns [3]; a momentum-unaware venue there would be repeatedly picked off [2, 4].

## 3.3 Depth

Depth at each step is derived deterministically from spread width, not simulated as an independent process:

$$\mathrm{depth}_t \;=\; \bar{D} \cdot \frac{\overline{S}}{S_t},$$

where $S_t$ is the full spread and $\overline{S}$ its session mean. The economic story is simple: when the external market maker is confident, they quote tight and large; when stressed, they pull size and widen. The hedge router on $A$ reads this array with the same 200 ms ($B$) / 170 ms ($C$) latency offset as the mid, and caps each hedge leg at the available depth.

## 3.4 One stock, two venues

$B$ and $C$ are two `Market` objects wrapping the same `Stock` via `copy.deepcopy`. The fundamental path $V_t$ is shared; the noise draws and spread realisations are independent. We deliberately do not model $B/C$ price divergence: the cross-venue edge in this project is the latency asymmetry (HFTs see $B/C$ at 50 ms while we see them at 200/170 ms), not price disagreement between venues.

## 3.5 Validation

Figure 2 overlays the four retained regimes on a shared GARCH-plus-jump run — the one fundamental where the models visibly differentiate. The top two panels plot ask and bid quotes for Static, Stochastic, Adaptive, and Skewed against the common mid; the four middle panels isolate each width path with a Pearson correlation to realized vol annotated in the title; the bottom panel shows the skew $\delta_t$ activating around the jump under the Skewed regime and remaining near zero elsewhere. The Static width is flat up to the mid-scaling factor, while Stochastic, Adaptive, and Skewed all co-move with $\widehat{RV}_t$ ($\rho \approx 1.00$, 1.00, 0.97), confirming the vol-adaptive mechanism fires as intended.

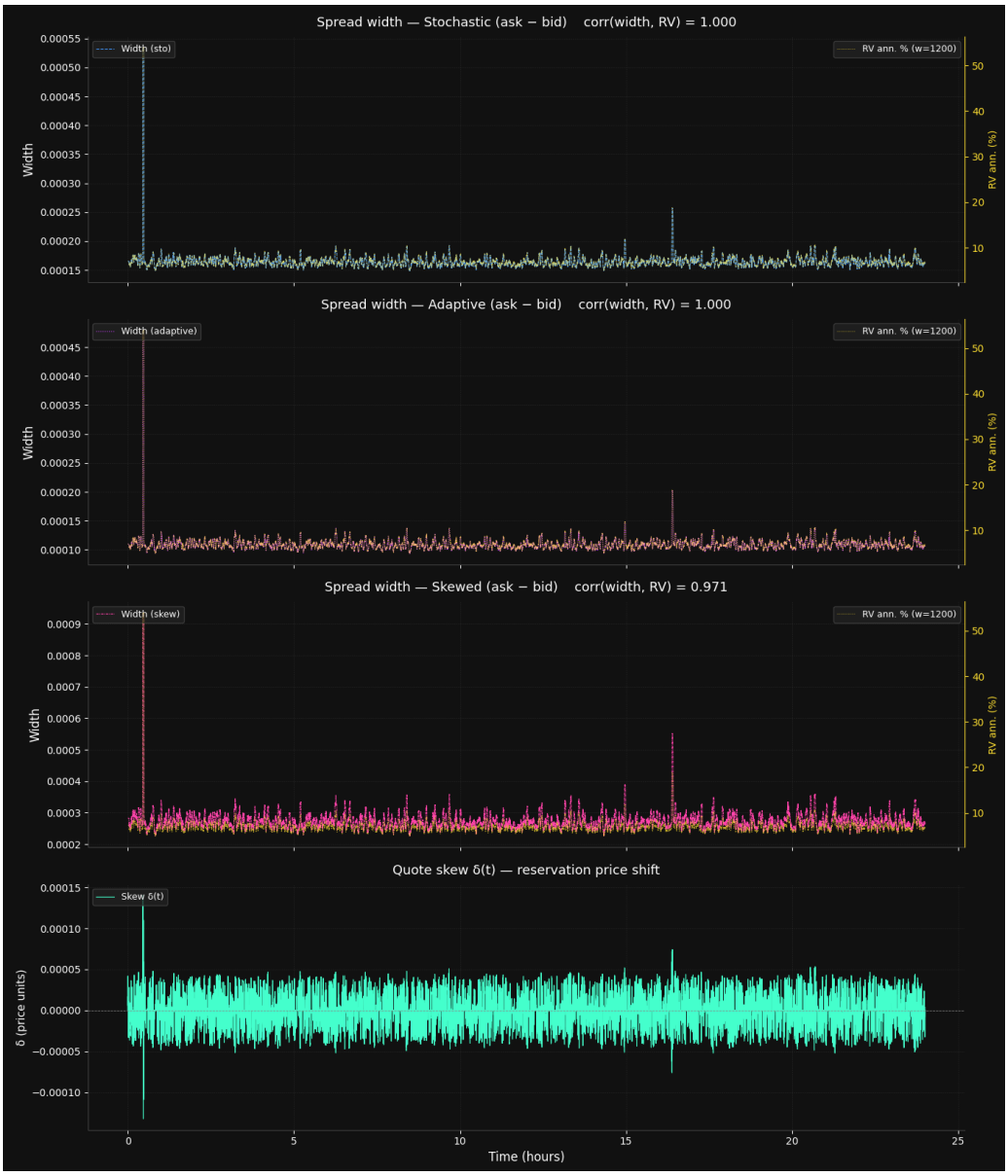



Figure 2: Ask and bid quotes, width panels with realized-vol overlay, and the Skew $\delta_t$ path on a shared GARCH+jump run.

# 4 Market-Making Logic on Exchange A

Exchange $A$ has no history, so every quote we post is derived from the state of $B$, $C$, our inventory $q$, and our capital $K$. Given these inputs the quoter must output two things each step: the set of orders to post and the set of resting orders to cancel. We describe here the modelling decisions that produce those two lists, not the Python that assembles them. The six subsections follow the order in which the decisions are composed: read the reference price, compute a reservation price, widen a spread around it, skew asymmetrically and lay down a ladder, decide whether to re-quote at all, and finally hedge when inventory breaches its limit.

## 4.1 Fair mid under latency asymmetry

We never quote against $A$'s own mid: $A$ has no informative price until we seed one. The reference is instead a cross-venue best constructed from $B$ and $C$ read with their respective latency offsets $\ell_B = 200\,\text{ms}$ and $\ell_C = 170\,\text{ms}$,

$$\text{bid}_t^{\text{fair}} = \max\big(\text{bid}_B(t-\ell_B),\ \text{bid}_C(t-\ell_C)\big), \qquad \text{ask}_t^{\text{fair}} = \min\big(\text{ask}_B(t-\ell_B),\ \text{ask}_C(t-\ell_C)\big),$$

with $\text{mid}_t^{\text{fair}} = \frac{1}{2}(\text{bid}_t^{\text{fair}} + \text{ask}_t^{\text{fair}})$. The choice of the cross-venue *best* rather than a volume-weighted average is a defence against being arbitraged: clients can always transact at the tighter side elsewhere, so anything wider on $A$ is free optionality handed to adversarial flow. HFTs read the same feeds at $\ell_{\text{HFT}} = 50\,\text{ms}$, so the price we quote against is always structurally stale relative to theirs; this gap is priced explicitly as a spread component in Section 4.3.

## 4.2 Reservation price: Avellaneda–Stoikov with infinite horizon

The classical inventory-shift mechanism of Avellaneda and Stoikov [1] uses a finite horizon $T$ with the penalty $\gamma\sigma^2(T-t)$. That formulation is inappropriate for a 24/5 FX market: there is no terminal liquidation time at which inventory becomes worthless. We replace $T-t$ by a stationary discount $1/\omega$ with $\omega = 1/(8\,\text{h})$ — one FX session as the effective risk horizon — which is the infinite-horizon limit of the same HJB system, studied formally by Guéant et al. [7].

A second correction is numerical. Applying the raw Avellaneda–Stoikov shift with inventory in EUR (range 0–900k) produces a reservation displacement of several hundred basis points at ten percent inventory, three orders of magnitude larger than the half-spread itself. We therefore work with the normalised ratio $\rho = q/K$ and express the shift as a fraction of the half-spread, so that at $|\rho| = 1$ the quoted centre is displaced by exactly one half-spread:

$$r_t \;=\; \text{mid}_t^{\text{fair}} \;-\; \rho \cdot h_t.$$

Direction is preserved (a long EUR position drags the centre down and attracts sells); magnitude is now calibrated to the spread itself, with no new free parameter beyond $\gamma$.

## 4.3 Three-component spread with fee floor

The half-spread $h_t$ is built as the sum of three economically distinct adverse-selection premia, floored by a fee-cover term,

$$h_t \;=\; \tfrac{1}{2}\big(h_t^{\text{AS}} + h_t^{\text{lat}} + h_t^{\text{inv}}\big), \qquad h_t \;\geq\; \phi \cdot \text{fee}_{\text{maker}}^{A} \cdot \text{mid}_t^{\text{fair}}.$$

The Avellaneda–Stoikov component

$$h_t^{\mathrm{AS}}(\mathrm{bps}) \;=\; \gamma \cdot (\sigma \cdot 100)^2 \cdot \tfrac{1}{\omega} \;+\; \tfrac{2}{\gamma}\,\ln\!\big(1+\gamma/k\big)$$

combines inventory-risk aversion (first term) with the convexity of the exponential order-arrival intensity (second term), as derived by Avellaneda and Stoikov [1] after the infinite-horizon substitution. The latency term $h_t^{\mathrm{lat}} = 2\sigma_{\mathrm{sec}}\sqrt{\Delta\ell}$, with $\Delta\ell = w_B(\ell_B - \ell_{\mathrm{HFT}}) + w_C(\ell_C - \ell_{\mathrm{HFT}})$, is two standard deviations of the expected adverse move during the window in which HFTs see the reference feeds and we do not — a cushion against the free option they hold against our stale quotes. The capacity term $h_t^{\mathrm{inv}} = \alpha_s \rho^2 h_t^{\mathrm{AS}}$ is quadratic in the inventory ratio: it is negligible near flat and widens sharply as we approach the delta limit, pushing flow away from the side that would deepen our position. Finally the floor $\phi \cdot \mathrm{fee}^A_{\mathrm{maker}} \cdot \mathrm{mid}_t^{\mathrm{fair}}$ guarantees that the inception spread covers the maker fee on every fill: no trade is ever gross-negative, only fee-negative offset by subsequent inventory P&L. The floor binds in calm regimes where the three adversarial premia would otherwise collapse below the fee.

## 4.4 Asymmetric skew and ladder

The reservation shift in Section 4.2 is symmetric: it moves both sides of the quote by the same amount. Guéant [6] shows that the asymptotic optimum is instead an *asymmetric* split of the half-spread — widen the side that would deepen inventory, tighten the side that would reduce it. We therefore apply the Guéant skew

$$\Delta_t \;=\; q \cdot \sqrt{\frac{\gamma\sigma^2}{2k}}, \qquad \mathrm{ask}_t^{\mathrm{post}} = r_t + h_t - \Delta_t, \qquad \mathrm{bid}_t^{\mathrm{post}} = r_t - h_t - \Delta_t,$$

with $\Delta_t$ clipped so the tighter side can never breach the fee floor.

The subject requires ten levels on each side. Levels step one tick further from the mid at each rung; sizes decay exponentially and are skewed against inventory,

$$\mathrm{size}_i^{\mathrm{bid}} \;=\; Q_0\, e^{-\beta i}\,(1 - \tfrac{1}{2}\overline{\rho}), \qquad \mathrm{size}_i^{\mathrm{ask}} \;=\; Q_0\, e^{-\beta i}\,(1 + \tfrac{1}{2}\overline{\rho}),$$

with $\overline{\rho} = \mathrm{clip}(\rho, -1, 1)$. This is a second, complementary inventory correction acting through *depth* rather than *price*: even if the Guéant skew is clipped by the fee floor, the ladder sizes still bias against further accumulation. Exponential decay concentrates displayed liquidity at the top of book, consistent with the empirical shape of limit-order-book queues.

## 4.5 Selective re-quoting

Re-quoting every step is expensive in cancel/post messages and costly in time priority: orders that are still valid lose their queue position on every spurious cancel. We use a strict six-priority hierarchy, with each lower priority firing only if none above it did (Table 1). Priority 2 treats the 08:00 and 16:00 UTC FX session boundaries as hard resets of the full ladder, so session-specific state does not accumulate across regimes. Priority 3 closes the price-drift leak; priority 5 closes a subtler silent-accumulation leak where market prices are flat but our own inventory has moved enough to invalidate the ladder skew. Priority 6 fires on pure staleness and re-quotes at a tighter 0.7× spread, so the refreshed quotes compete back into relevance rather than being re-posted at the same passive distance.

| P | Trigger | Action |
|---|---|---|
| 1 | First call | Full ladder, no cancels |
| 2 | Session boundary (08:00/16:00 UTC) | Cancel all, full ladder |
| 3 | $\lvert$best-price drift$\rvert > 0.25$ spread | Cancel all, full ladder |
| 4 | Fills / top-ups pending | Rebuild filled slots only |
| 5 | $\lvert\Delta q\rvert/K > 5\%$ since last reprice | Cancel all, full ladder |
| 6 | Order older than $\tau_{\text{stale}}$ | Cancel stale, reprice at $0.7\times$ spread |

Table 1: Requote priorities; at most one rule fires per step.

## 4.6 Hedge router

When either $|q| > \delta_{\text{lim}} \cdot K/2$ or $|q \cdot \text{mid}^{\text{fair}}| > \delta_{\text{lim}} \cdot K/2$ — that is, either the EUR leg or its USD notional breaches the delta limit on half-capital — we externalise risk through taker market orders on $B$ and $C$. Three decisions shape the router. First, the USD notional of EUR inventory is valued at the $B/C$ cross-venue mid, not at $\text{mid}_A$, because $A$'s mid is polluted by our own skewed quotes. Second, the split between venues is depth-aware: each venue is scored by

$$s_v \;=\; \frac{1}{\text{fee}^v_{\text{taker}} \;+\; \sigma_{\text{sec}} \cdot \ell_v \;+\; \lambda \cdot \text{mid}^{\text{fair}}/D_v},$$

combining the taker fee, the expected adverse move during execution lag, and a market-impact penalty inversely proportional to the available depth $D_v$ at the lagged step. The hedge is split $s_B : (s_B + s_C)$ with overflow re-routed to the other venue when the preferred one is depth-constrained. Third, when post-hedge inventory remains above the tighter threshold hedge_partial_limit$\cdot K/2$ — meaning the combined $B$+$C$ depth was insufficient to bring us back — we flip an emergency flag that multiplies the Avellaneda–Stoikov inventory-risk term by $5\times$ on the next step. Our own quotes on $A$ become so aggressively asymmetric that organic client flow cleans up the residual. This is the Guéant et al. [7] inventory-penalty mechanism used as an overflow valve: we deliberately do not chase depth we cannot reach.

# 5 Client Order Flow on Exchange A

The quoter of Section 4 sets prices, but the only channel through which those prices become realised revenue is the stream of client orders reaching Exchange $A$. We therefore need a generative model of that flow with three layers: *when* an order arrives, *which type* it is, and *how large* it is. The intensity baseline $A$, the price sensitivity $k$, the tail index $\alpha$ of order sizes and the market-order mix $p_{\mathrm{MO}}$ are design parameters of the simulator: they are held fixed across Phase 1 and Phase 2 and are not re-estimated by the calibration pipeline of Section 8. What Phase 2 recalibrates are the quoter's own risk parameters — the client-flow knobs above remain exogenous throughout.

## 5.1 Poisson arrivals with price-sensitive intensity

Buy-side and sell-side clients are modelled as two independent inhomogeneous Poisson processes whose instantaneous rates depend on how far our posted quote sits from the prevailing mid on the relevant side. Tightening the ask attracts buy-side flow without mechanically also attracting sell-side flow, which matches the empirical asymmetry of electronic FX and preserves the separation of controls the quoter relies on. With the mid $m_t$ and our top-of-book $(\mathrm{bid}_t, \mathrm{ask}_t)$, the per-side distances in basis points are

$$\delta_t^{\mathrm{ask}} = 10^4 \, \frac{\mathrm{ask}_t - m_t}{m_t}, \qquad \delta_t^{\mathrm{bid}} = 10^4 \, \frac{m_t - \mathrm{bid}_t}{m_t},$$

and the arrival intensities take the exponential kernel of Avellaneda and Stoikov [1],

$$\lambda_t^{\mathrm{buy}} \;=\; A \, \exp\!\big(-k \, \delta_t^{\mathrm{ask}}\big), \qquad \lambda_t^{\mathrm{sell}} \;=\; A \, \exp\!\big(-k \, \delta_t^{\mathrm{bid}}\big).$$

Over a simulation step of length $\Delta t$ the number of arrivals on each side is drawn independently as $N^{\mathrm{side}} \sim \mathrm{Poisson}(\lambda_t^{\mathrm{side}} \cdot \Delta t)$, so a step can produce zero, one, or several orders per side. With the defaults $A = 7 \cdot 10^{-3}\,\mathrm{s}^{-1}$ and $k = 0.3\,\mathrm{bp}^{-1}$, the half-life of the intensity is $\ln(2)/k \approx 2.3\,\mathrm{bp}$: a quote posted two bps outside the fair mid already attracts roughly half the arrivals that a touch-priced quote would.

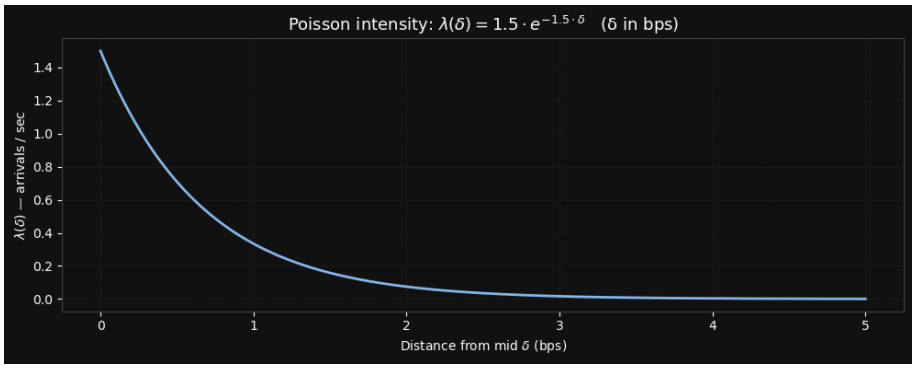



Figure 3: Arrival intensity $\lambda(\delta) = A\,e^{-k\delta}$ as a function of the distance from the fair mid, with $A = 0.007\,\mathrm{s}^{-1}$ and $k = 0.3\,\mathrm{bp}^{-1}$. Each bp of extra distance costs the quote approximately 26% of its arrival rate.

## 5.2 Market-versus-limit mix and limit placement

Each arrival is independently classified market or limit by a Bernoulli trial with probability $p_{\mathrm{MO}} = 0.5$. Market orders cross immediately at the touch — buys at $\mathrm{ask}_t$, sells at $\mathrm{bid}_t$ —

and are the mechanism through which the quoter earns its inception spread. Limit orders rest inside the book at a random offset from the mid,

$$p^{\mathrm{LO,\,buy}} \;=\; m_t\left(1-\tfrac{\varepsilon}{10^4}\right), \qquad p^{\mathrm{LO,\,sell}} \;=\; m_t\left(1+\tfrac{\varepsilon}{10^4}\right), \qquad \varepsilon \sim \mathrm{Exp}(k).$$

Using the same $k$ in both the arrival intensity and the limit-placement distribution is a self-consistency choice rather than a fit: clients who are dissuaded from crossing at rate $k$ are assumed to post passively at the same rate $k$. This keeps the flow symmetric under a shift of the quoted distance — widening our spread by $\Delta\delta$ reduces arrivals by $e^{-k\Delta\delta}$ and simultaneously pushes the average passive limit deeper by the same factor, with no second free parameter to tune.

## 5.3 Heavy-tailed order sizes

Client notional sizes are drawn from a truncated Pareto on the interval $[x_{\min}, x_{\max}] = [10^3, 10^5]$ EUR with tail index $\alpha = 1.5$, in line with the power-law volume distribution of Gabaix et al. [5]. Truncation is not only empirical realism — the exchange rejects orders outside its permitted notional band — but also a numerical safeguard: an untruncated Pareto with $\alpha = 1.5$ has infinite variance, and Monte-Carlo P&L moments would be dominated by a handful of outliers instead of by the structural behaviour of the quoter. The truncated inverse-CDF sampler is

$$X \;=\; x_{\min}\Big(1 - U\big(1-(x_{\min}/x_{\max})^{\alpha}\big)\Big)^{-1/\alpha}, \qquad U \sim \mathcal{U}(0,1),$$

which respects the bounds exactly without rejection sampling. The resulting distribution concentrates mass near $x_{\min}$ but assigns a non-trivial weight to the largest permitted ticket, so a typical session mixes many small fills with a few that can move the inventory by several percent of capital in a single trade.

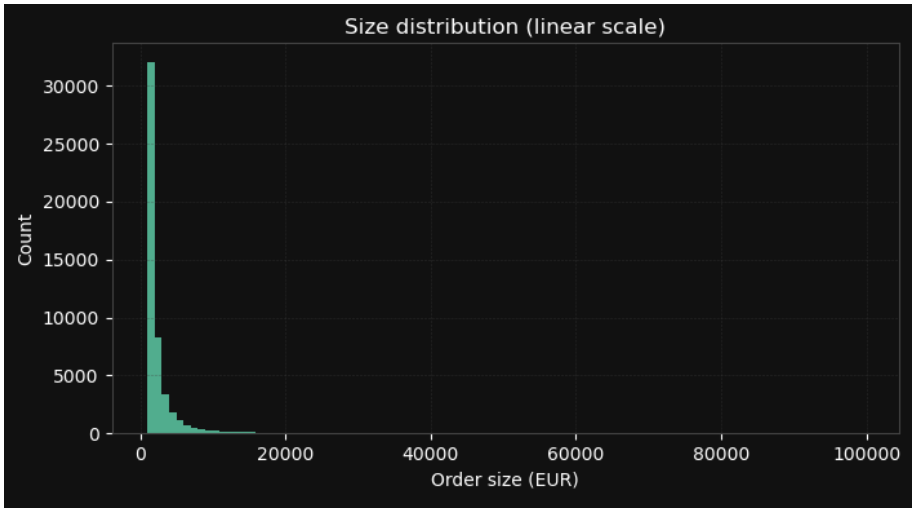



Figure 4: Truncated Pareto size distribution on $[10^3, 10^5]$ EUR with $\alpha = 1.5$: mass concentrates near the lower bound while the heavy tail keeps a non-trivial weight at the hard cap $x_{\max}$.

## 5.4 Consumption by the order book

Each generated order is tagged with origin = `client` and pushed into the same matching engine that ingests our own quotes (Section 2). A client market buy walks the ask ladder level by level under price–time priority, partially filling our resting quotes until its size is exhausted; a client limit order either crosses and generates an immediate fill at the touch we posted or rests in the book, where it remains eligible to be matched against a future quote. The matching rule restricts trades to client-versus-market-maker pairs, so client-to-client crossings never occur and the realised fill price is always our passive quote. This closed loop — client intensity shaped by our quote, fills generated only through our ladder — is precisely the structure assumed by the Avellaneda–Stoikov and Guéant et al. [7] half-spread derivations, and it makes the intensity baseline $A$ and decay $k$ the two knobs that jointly control fill rate and adverse-selection severity alongside our posted half-spread $h_t$.

# 6 Inventory and P&L Tracking

Every fill on $A$ and every hedge leg on $B/C$ is appended to a single trade log with columns $(t, \text{side}, p, x, m^{\text{fair}}, \text{cf}, \phi, \text{is_hedge}, q_{\text{after}})$, where cf is the signed USD cash-flow and $\phi$ the fee paid on that leg. All P&L figures reported in the backtest (Section 7) are derived from this log.

## 6.1 Sign and fee convention

Cash-flows are signed from the market-maker's perspective in USD: positive when cash flows in, negative when it flows out. For a sell fill on $A$ at price $p$ and size $x$ (EUR), the cash-flow is $+p\,x\,(1-\phi_A^{\text{mak}})$; for a buy fill it is $-p\,x\,(1+\phi_A^{\text{mak}})$. Hedge legs cross the opposite side on $B$ or $C$ and pay the corresponding taker fee. Fees are therefore *already embedded* in cf. We log $\phi$ separately only to isolate the fee line in the decomposition below — it is not subtracted a second time.

## 6.2 Realized versus mark-to-market

At any time $t$ we split the total economic P&L into what is locked in and what remains at risk:

$$\text{MtM}_t \;=\; \underbrace{\sum_{i:\,t_i\le t} \text{cf}_i}_{\text{realized}} \;+\; \underbrace{q_t \cdot m_t^{\text{fair}}}_{\text{unrealized}}, \tag{1}$$

where $q_t$ is the open EUR inventory after the last fill and $m_t^{\text{fair}}$ is the cross-venue fair mid of Section 4.1. Valuing inventory at the fair mid rather than at $A$'s own noised mid removes the venue-specific microstructure noise from the valuation: the unrealized line reflects only the economic value of the EUR position, not the idiosyncratic distortion of the venue on which it sits.

## 6.3 Inception spread versus mid-to-mid revaluation

Equation (1) separates *what is settled* from *what is still open*, but it does not say *why* the P&L got to its current value. For that we use a second, orthogonal, decomposition:

$$\text{MtM}_t \;=\; \underbrace{\sum_{i\in\mathcal{M},\,t_i\le t} s_i\,(p_i - m_{t_i}^{\text{fair}})\,|x_i|}_{\text{inception spread } \mathcal{S}_t} \;+\; \underbrace{\text{Reval}_t}_{\text{mid-to-mid revaluation}} \;-\; \underbrace{\mathcal{F}_t}_{\text{fees}}, \tag{2}$$

with $\mathcal{M}$ the set of MM fills on $A$ (hedge legs excluded), $s_i = +1$ for a sell and $-1$ for a buy, and $\mathcal{F}_t = \sum_{i\le t} |\phi_i|$ the cumulative fees. $\mathcal{S}_t$ is the spread we captured *at the moment* of each fill — it is the pure market-making edge, measured before any subsequent market move. $\text{Reval}_t$ is the residual, defined by (2), and isolates the P&L generated purely by $m^{\text{fair}}$ drifting after a fill on the open position it left us with; it is a directional, non-market-making component. This is precisely the distinction between inception spread and mid-to-mid revaluation that the brief asks for.

## 6.4 Continuous MtM and final inventory

Evaluating (1) only at fill events gives a sparse view. For the backtest report we reconstruct MtM on the full simulation grid: the cumulative cash, fees, and inception spread columns are forward-filled onto every step, and residual inventory is marked at the concurrent fair mid. This yields a continuous curve plotted alongside the discrete realized-cash ladder in Section 7 and makes the mid-to-mid revaluation component visible between fills, not just at them. At horizon $T$ we close the run with the triple $(\text{realized}_T,\ q_T \cdot m_T^{\text{fair}},\ q_T)$: the first is the locked-in USD P&L, the second is the final inventory value in USD — the mark at which we would liquidate the remaining EUR — and the third is the raw EUR position we carry out.

# 7 Phase 1 Backtest Report

This section addresses Subject Task 7 for the Phase 1 quoter: a 30 day, $\Delta t$=0.1 s simulation of the heuristic market maker (default `QuoterConfig`, no calibration). All figures reproduced here are taken directly from `ctrl_p1.report()` in `test/phase2_calibration.ipynb` (cell 5).

## 7.1 Simulation setup

The quoter is initialised with the *heuristic* defaults summarised in Table 2. No parameter has been estimated from data at this stage — risk aversion, depth decay and the inventory limits are chosen by reference to Avellaneda–Stoikov / Guéant / Cartea–Jaimungal rather than calibrated to the simulated book. Capital is \$1 M split 50/50 between EUR and USD at inception.

| Parameter | Value |
|---|---|
| Capital $K$ | \$1 000 000 |
| $\gamma$ (risk aversion) | 0.10 |
| $k$ (arrival intensity) | 0.30 |
| $\omega$ (discount rate) | $3.47\times10^{-5}\,\mathrm{s}^{-1}$ ($\approx$ 8 h) |
| $Q_{\text{base}}$ | 70 000 EUR |
| Ladder levels / $\beta$ | 10 / 0.30 |
| `delta_limit` | $0.90\,(K/2) = \pm450$ k |
| `hedge_partial_limit` | $0.80\,(K/2) = \pm400$ k |
| `vol_window_s` | 60 s |
| Fees A / B / C (maker, taker) | 1/4, 1/2, 1/3 bps |
| Latencies B / C / HFT | 200 / 170 / 50 ms |

Table 2: Phase 1 heuristic configuration (defaults of `QuoterConfig`).

## 7.2 Aggregate results

Table 3 reports the 30-day P&L decomposition produced by `PnLTracker.report`. Total mark-to-market P&L is positive (+\$7 484), but the breakdown is informative: the inception spread captured at fill time accounts for +\$8 415, slightly more than the total; unrealised inventory revaluation contributes +\$3 643 on top; fees cost \$4 575, of which \$4 030 are maker fees on exchange A.

The large gap between realised (−\$140 k) and unrealised (+\$147 k) P&L reflects a persistently open EUR leg: the final net inventory of +137 k EUR represents 13.7% of capital at expiry. The realised P&L is not the spread revenue but the cash cost of accumulating that open position; the unrealised P&L is the mark of that position at the terminal mid.

## 7.3 Market prices — three venues (Subject 7.1–7.2)

Figure 5 stacks best bid, best ask and mid for the three venues plus a spread-in-bps panel at the bottom. EUR/USD drifts from 1.100 down to $\approx$ 1.058 in the first third of the month then partially recovers to $\approx$ 1.076. Market A (the exchange we quote) is visibly

| Metric | USD |
|---|---|
| **Total MtM P&L** | **+7 483.54** |
| Realised cash P&L | −139 816.52 |
| Unrealised (open inv. × mid) | +147 300.06 |
| Inception spread P&L | +8 415.27 |
| Inventory revaluation P&L | +3 642.96 |
| Total fees paid | 4 574.69 |
| Maker fees (exchange A) | 4 030.48 |
| Taker fees (hedge B/C) | 544.21 |
| MM fills | 14 375 (479/day) |
| Hedge legs | 10 |
| Final net inventory | +136 838 EUR |
| Spread A (MM quoted) | 5.94 bps ($\sigma = 0.45$) |
| Spread B / C (reference) | 2.50 / 1.41 bps |

Table 3: Phase 1 aggregate KPIs (30 days).

wider than the references B (75% volume, 200 ms latency) and C (25% volume, 170 ms latency) — consistent with the quoter pricing in its own risk and the HFT latency edge.

## 7.4 Top 10 MM fills (Subject 7.2)

Figure 6 highlights the ten largest MM fills on market A, with marker area proportional to EUR notional and annotations giving {size, captured inception spread in bps}. Fills are clustered in the 52–59 k EUR range (all partials of a single quoted clip), and the captured per-trade inception spread ranges from 0.0 to 1.7 bps — a small fraction of the 5.94 bps half-quoted spread, the remainder being the MM's own risk buffer rather than realised revenue.

The per-venue version on B and C is omitted: the MM never posts passive quotes on those venues and only ten hedge legs fire in the whole month, so there is no meaningful top-10 to show.

## 7.5 Inventory vs. EUR/USD (Subject 7.3)

Figure 7 overlays the fair cross-venue mid and the absolute EUR leg (capital half is 500 k, so inventory is the distance from that baseline). Two observations stand out. First, the inventory trace is *anti-correlated* with the mid over most of the month — the quoter correctly accumulates EUR on dips and unwinds it on rallies. Second, the `delta_limit` band at ±450 k is never hit, yet the absolute EUR leg routinely exceeds 800 k (i.e. net inventory ≈ ±300 k): the bands are too wide to trigger hedging at the frequency one would expect for a market maker, which is why only ten hedge legs fire in 30 days.

## 7.6 Per-trade MtM P&L percentiles (Subject 7.4)

Figure 8 shows the distribution of per-trade mark-to-market P&L as a function of $\tau$, the time elapsed since the trade's inception, estimated on a 500-fill sample. The 5th–95th percentile band is roughly ±\$3 500 at $\tau = 0$ and stays at that width for the full 30 days, which is consistent with the variance of the open-inventory leg on EUR/USD rather than

Figure 5: Phase 1: best bid / ask / mid for markets A, B, C, plus instantaneous bid–ask spread (bps). 30 days.

with a convergent MM P&L. The mean sits slightly below zero at every $\tau$, indicating mild adverse selection on average — the spread captured on each fill is smaller than the subsequent mid drift against us.

## 7.7 Fill-rate analysis (Subject 7.5)

The matching engine records 14 375 MM fills over the month, averaging 479 fills / day at an average clip size of 2 616 EUR. The breakdown by ladder level (Table 4) shows that level 1 absorbs 99.6% of executions, level 2 barely 0.4%, and levels 3–10 are essentially idle. Only 0.5% of orders are fully filled — the remaining 99.5% are partials, the size at the touch being systematically larger than client clip sizes.

A caveat on the `overall fill rate` line of the printed report: the quotes-posted counter is currently not incremented, so the overall rate prints as 0%. This is a reporting bug, not an execution result; the absolute fill counts and per-level breakdown above are

Figure 6: Top 10 MM fills on market A, by EUR size. Green = buy, red = sell. The top fills are split roughly 7 sells / 3 buys, mirroring the price down-leg then partial recovery visible in Figure 5.

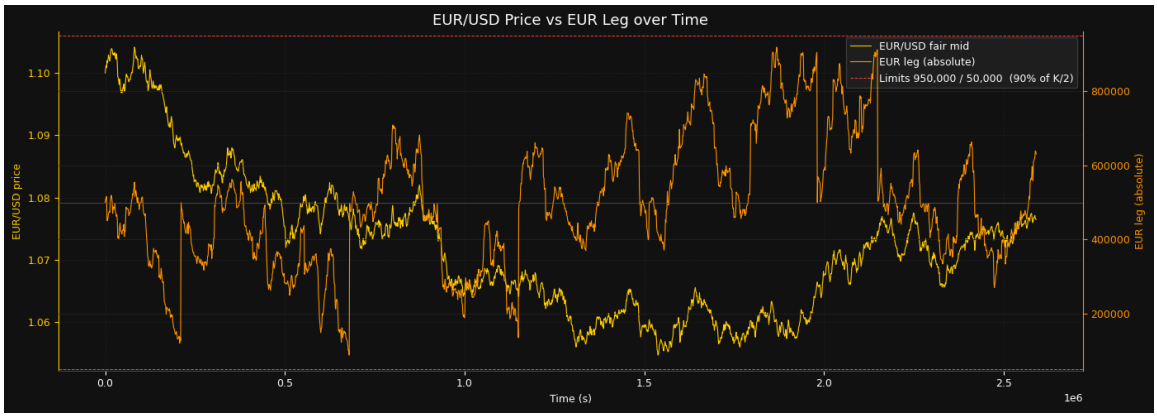



Figure 7: EUR/USD fair mid (blue) against absolute EUR leg (orange), with `delta_limit` bands.

unaffected.

## 7.8 Interpretation

Phase 1 behaves as a reasonable baseline. The pipeline captures spread at every fill (+\$8 415 of inception spread P&L), fees are contained at roughly 0.46 bps of notional, the inventory trace in Figure 7 has the correct anti-correlation with the mid, and the quoted spread on A (5.94 bps) is plausible given that the reference venues B and C trade 2.50 and 1.41 bps wide and that HFTs are 150 ms ahead of us on the references. Nevertheless, three structural weaknesses emerge that the calibration step is meant to correct.

1. **P&L is sensitive to the direction of EUR.** Realised (−\$140 k) and unrealised (+\$147 k) P&L nearly cancel, so the positive headline MtM is largely the result of EUR drifting upward over the window. A mirror path would have flipped the sign.

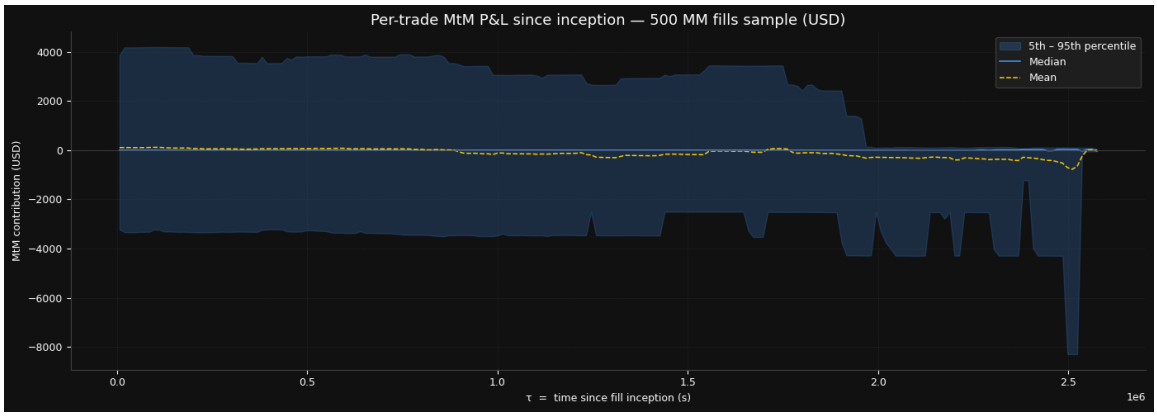



Figure 8: Per-trade MtM contribution (USD) vs. time since fill inception. Shaded band: 5th–95th percentile; solid = median; dashed = mean. Sample of 500 MM fills.

| Level | Fill count | Relative rate |
|---|---|---|
| 1 | 14 319 | 1.000 |
| 2 | 54 | 0.004 |
| 3 | 2 | 0.000 |
| 4–10 | 0 | 0.000 |

Table 4: Phase 1 fills by ladder level. The 10-level quote ladder is substantially over-provisioned for the simulated arrival intensity.

2. **Hedging is effectively off.** At `delta_limit` = 0.90 and `hedge_partial_limit` = 0.80, only 10 hedge legs fire in 30 days while the EUR leg swings 0 – 900 k in absolute value. Tighter bands (the reference demo uses 0.60 / 0.05) would bring hedging online.

3. **Per-fill capture is thin.** Figure 6 shows only 0.0–1.7 bps captured on the largest fills against a quoted 5.94 bps. Most of the quoted half-spread is a risk buffer that does not convert into revenue; the mild but persistent negative mean in Figure 8 suggests adverse selection is part of the story.

## 7.9 Take-away for Phase 2

The results motivate the direction taken by the calibration step in Phase 2: $\gamma$ is recalibrated from 0.10 down to $\approx 0.008$ and $Q_{\text{base}}$ from 70 k down to $\approx 7.4$ k EUR, both of which should make quotes more reactive to inventory and better matched to the simulated arrival sizes. The fill-rate ladder usage and the hedging frequency in particular will be the diagnostics to watch once the calibrated configuration is deployed on a fresh price path.

# 8 Phase 2 — Calibration from Trading History

# 9 Phase 3 — Fallback Liquidity Under HFT Competition

# 10 Conclusion

# References


[1] Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.

[2] Bruno Biais, Pierre Hillion, and Chester Spatt. An empirical analysis of the limit order book and the order flow in the Paris bourse. *The Journal of Finance*, 50(5): 1655–1689, 1995.

[3] Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014.

[4] Robert F. Engle and Andrew J. Patton. Impacts of trades in an error-correction model of quote prices. *Journal of Financial Markets*, 7(1):1–25, 2004.

[5] Xavier Gabaix, Parameswaran Gopikrishnan, Vasiliki Plerou, and H. Eugene Stanley. A theory of power-law distributions in financial market fluctuations. *Nature*, 423 (6937):267–270, 2003.

[6] Olivier Guéant. Optimal market making. *Applied Mathematical Finance*, 24(2):112–154, 2017.

[7] Olivier Guéant, Charles-Albert Lehalle, and Joaquin Fernandez-Tapia. Dealing with the inventory risk: A solution to the market making problem. *Mathematics and Financial Economics*, 7(4):477–507, 2013.

[8] Richard Roll. A simple implicit measure of the effective bid-ask spread in an efficient market. *The Journal of Finance*, 39(4):1127–1139, 1984.